Equal real Eigenvalues with multiplicity three

An **Eigenvalue with multiplicity three** is an Eigenvalue that shows up three times in the characteristic equation. Finding the general solution in this case is just an extension of what we learned to do with Eigenvalues of multiplicity two.

The multiplicity we find for the Eigenvalues from the characteristic equation is the **algebraic multiplicity**, while the number of Eigenvectors we find for each Eigenvalue is that Eigenvalue's **geometric multiplicity**.

Given a 3×3 matrix, if we find one Eigenvalue with multiplicity three, and then three linearly independent Eigenvectors for that Eigenvalue, then the algebraic and geometric multiplicity are equal, and the solution to the system will be just like the solution for distinct real Eigenvalues,

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + \ldots + c_n\overrightarrow{x_n}.$$

Eigenvalues with a defect

When we have a 3×3 matrix where we find one Eigenvalue with multiplicity three that produces fewer than three linearly independent Eigenvectors, the difference between the algebraic and geometric multiplicity is the **defect**.

When we have an Eigenvalue with multiplicity three with a defect of two (we found only one linearly independent Eigenvector), we'll still use the first Eigenvector $\overrightarrow{k_1}$ for the first solution $\overrightarrow{x_1}$, but then we'll need these special equations,

© Krista King Math



$$(A - \lambda_1 I)\vec{p_1} = \vec{k_1}$$

$$(A - \lambda_1 I)\vec{q_1} = \vec{p_1}$$

to build the three Eigenvectors,

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t}$$

$$\vec{x_2} = \vec{k_1} t e^{\lambda_1 t} + \vec{p_1} e^{\lambda_1 t}$$

$$\vec{x_3} = \vec{k_1} \frac{t^2}{2} e^{\lambda_1 t} + \vec{p_1} t e^{\lambda_1 t} + \vec{q_1} e^{\lambda_1 t}$$

that we can plug into the general solution equation.

But when we have an Eigenvalue with multiplicity three with a defect of one (we found only two linearly independent Eigenvectors), then we'll use the first two Eigenvectors $\vec{k_1}$ and $\vec{k_2}$ in this special equation,

$$(A - \lambda_1 I)\vec{p_1} = a_1\vec{k_1} + a_2\vec{k_2}$$

to build the third solution vector

$$\vec{x_3} = \vec{p_1} e^{\lambda_1 t} + (a_1\vec{k_1} + a_2\vec{k_2}) t e^{\lambda_1 t}$$

Let's do an example with an equal real Eigenvalue with multiplicity three and a defect of one, where the Eigenvalue with multiplicity three produces only two linearly independent Eigenvectors.

Example

Find the general solution to the system of differential equations.

© Krista King Math



$$x_1'(t) = x_1$$

$$x_2'(t) = x_1 + 2x_2 + x_3$$

$$x_3'(t) = -x_1 - x_2$$

We can represent this system in matrix form as

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix} \vec{x}$$

We'll need to find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 0 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & -\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & -\lambda \end{vmatrix}$$

© Krista King Math

$$|A-\lambda I| = (1-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ -1 & -\lambda \end{vmatrix} - 0\begin{vmatrix} 1 & 1 \\ -1 & -\lambda \end{vmatrix} + 0\begin{vmatrix} 1 & 2-\lambda \\ -1 & -1 \end{vmatrix}$$

$$|A-\lambda I| = (1-\lambda)[(2-\lambda)(-\lambda)-(1)(-1)]$$

$$-0[(1)(-\lambda)-(1)(-1)]+0[(1)(-1)-(2-\lambda)(-1)]$$

$$|A-\lambda I| = (1-\lambda)(-2\lambda+\lambda^2+1)$$

$$|A-\lambda I| = -2\lambda+\lambda^2+1+2\lambda^2-\lambda^3-\lambda$$

$$|A-\lambda I| = 1-3\lambda+3\lambda^2-\lambda^3$$

Solve the characteristic equation for the Eigenvalues.

$$1-3\lambda+3\lambda^2-\lambda^3=0$$

$$\lambda^3-3\lambda^2+3\lambda-1=0$$

$$(\lambda-1)(\lambda-1)(\lambda-1)=0$$

Then for these Eigenvalues, $\lambda_1=\lambda_2=\lambda_3=1$, we find

$$A-1I=\begin{bmatrix} 1-1 & 0 & 0 \\ 1 & 2-1 & 1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$A-1I=\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

© Krista King Math



If we turn this first row into an equation, we get

$$k_1 + k_2 + k_3 = 0$$

$$k_1 = -k_2 - k_3$$

We'll choose $(k_2, k_3) = (-1, 0)$, which results in $k_1 = 1$.

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

and therefore

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{t}$$

But we can actually find a second linearly independent Eigenvector from $k_1 = -k_2 - k_3$. If we choose $(k_2, k_3) = (0, -1)$ instead, we get $k_1 = 1$, and we find a second Eigenvector.

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

and therefore

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{t}$$

© Krista King Math



There's no third vector we can find that won't be a linearly combination of these first two Eigenvectors, which means the Eigenvalue $\lambda = 1$ has a defect of one. We'll use the special formula for this case,

$$(A - \lambda_1 I)\overrightarrow{p_1} = a_1\overrightarrow{k_1} + a_2\overrightarrow{k_2}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ -a_1 \\ 0 \end{bmatrix} + \begin{bmatrix} a_2 \\ 0 \\ -a_2 \end{bmatrix}$$

Expanding this as a system of equations gives

$$0 = a_1 + a_2$$

$$p_1 + p_2 + p_3 = -a_1$$

$$-p_1 - p_2 - p_3 = -a_2$$

We can choose any values for $\overrightarrow{p_1}$, so let's choose something simple like $\overrightarrow{p_1} = (p_1, p_2, p_3) = (1,1,1)$. Then the system of equations becomes

$$0 = a_1 + a_2$$

$$3 = -a_1$$

$$-3 = -a_2$$

or

© Krista King Math



$$a_1 = -3$$

$$a_2 = 3$$

Then the third solution vector will be

$$\vec{x_3} = \vec{p_1}e^{\lambda_1 t} + (a_1\vec{k_1} + a_2\vec{k_2})te^{\lambda_1 t}$$

$$\vec{x_3} = \begin{bmatrix}1\\1\\1\end{bmatrix} e^t + \left(-3\begin{bmatrix}1\\-1\\0\end{bmatrix} + 3\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) te^t$$

$$\vec{x_3} = \begin{bmatrix}1\\1\\1\end{bmatrix} e^t + \left(\begin{bmatrix}-3\\3\\0\end{bmatrix} + \begin{bmatrix}3\\0\\-3\end{bmatrix}\right) te^t$$

$$\vec{x_3} = \begin{bmatrix}1\\1\\1\end{bmatrix} e^t + \begin{bmatrix}-3\\3\\0\end{bmatrix} te^t + \begin{bmatrix}3\\0\\-3\end{bmatrix} te^t$$

Therefore, the general solution will be

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1\begin{bmatrix}1\\-1\\0\end{bmatrix} e^t + c_2\begin{bmatrix}1\\0\\-1\end{bmatrix} e^t + c_3\left(\begin{bmatrix}1\\1\\1\end{bmatrix} e^t + \begin{bmatrix}-3\\3\\0\end{bmatrix} te^t + \begin{bmatrix}3\\0\\-3\end{bmatrix} te^t\right)$$

Now let's do an example with equal real Eigenvalues, but where we find only one Eigenvector associated with the three equal Eigenvalues.

Example

© Krista King Math



Find the general solution to the system of differential equations.

$$x_1'(t) = 2x_1 + x_2 - 6x_3$$

$$x_2'(t) = 2x_2 + 5x_3$$

$$x_3'(t) = 2x_3$$

We can represent this system in matrix form as

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 2 & 1 & -6 \\ 0 & 2 & 5 \\ 0 & 0 & 2 \end{bmatrix} \vec{x}$$

We'll need to find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 2 & 1 & -6 \\ 0 & 2 & 5 \\ 0 & 0 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 & 1 & -6 \\ 0 & 2 & 5 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 1 & -6 \\ 0 & 2-\lambda & 5 \\ 0 & 0 & 2-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

© Krista King Math

$$|A-\lambda I| = \begin{vmatrix} 2-\lambda & 1 & -6 \\ 0 & 2-\lambda & 5 \\ 0 & 0 & 2-\lambda \end{vmatrix}$$

$$|A-\lambda I| = (2-\lambda)\begin{vmatrix} 2-\lambda & 5 \\ 0 & 2-\lambda \end{vmatrix} + \begin{vmatrix} 0 & 5 \\ 0 & 2-\lambda \end{vmatrix} - 6\begin{vmatrix} 0 & 2-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|A-\lambda I| = (2-\lambda)((2-\lambda)(2-\lambda)-(5)(0))$$

$$+((0)(2-\lambda)-(5)(0))-6((0)(0)-(2-\lambda)(0))$$

$$|A-\lambda I| = (2-\lambda)(2-\lambda)(2-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(2-\lambda)(2-\lambda)(2-\lambda)=0$$

$$\lambda = 2,\ 2,\ 2$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = 2$, we find

$$A-2I = \begin{bmatrix} 2-2 & 1 & -6 \\ 0 & 2-2 & 5 \\ 0 & 0 & 2-2 \end{bmatrix}$$

$$A-2I = \begin{bmatrix} 0 & 1 & -6 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 0 & 1 & -6 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

© Krista King Math



$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn these first two rows into equations, we get $k_2 = 0$ and $k_3 = 0$. We can choose any value for k_1. If we choose $k_1 = 1$, we find

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Other than the trivial solution $\overrightarrow{k_1} = (0,0,0)$, any other (k_1, k_2, k_3) set we find from $(k_1, k_2, k_3) = (k_1,0,0)$ will result in a vector that's linearly dependent with the $\overrightarrow{k_1} = (1,0,0)$ vector we already found. So we can say that the Eigenvalue $\lambda_1 = \lambda_2 = \lambda_3 = 2$ produces only one Eigenvector.

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{2t}$$

Because we only find one Eigenvector for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 2$, we have to use $\overrightarrow{k_1} = (1,0,0)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 1 & -6 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$p_2 - 6p_3 = 1$$

© Krista King Math

$$5p_3 = 0$$

Using $p_3 = 0$ in the first equation gives $p_2 = 1$. We can choose any value for p_1, so we'll pick $p_1 = 1$ to get

$$\overrightarrow{p_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Then our second solution will be

$$\overrightarrow{x_2} = \overrightarrow{k_1} te^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

Now we'll use $\overrightarrow{k_1} = (1,0,0)$ and $\overrightarrow{p_1} = (1,1,0)$ to find a third solution.

$$(A - \lambda_1 I)\overrightarrow{q_1} = \overrightarrow{p_1}$$

$$\begin{bmatrix} 0 & 1 & -6 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$q_2 - 6q_3 = 1$$

$$5q_3 = 1$$

Using $q_3 = 1/5$ in the first equation gives $q_2 = 11/5$. We can choose any value for q_1, so we'll pick $q_1 = 1$ to get

© Krista King Math



$$\overrightarrow{q_1} = \begin{bmatrix} 1 \\ \frac{11}{5} \\ \frac{1}{5} \end{bmatrix}$$

Ideally, we'd like to simplify the solution by eliminating fractions, so we'll multiply through $\overrightarrow{q_1}$ by 5 to rewrite it as

$$\overrightarrow{q_1} = \begin{bmatrix} 5 \\ 11 \\ 1 \end{bmatrix}$$

Then our third solution will be

$$\overrightarrow{x_3} = \overrightarrow{k_1}\frac{t^2}{2}e^{\lambda_1 t} + \overrightarrow{p_1}te^{\lambda_1 t} + \overrightarrow{q_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\frac{t^2}{2}e^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}te^{2t} + \begin{bmatrix} 5 \\ 11 \\ 1 \end{bmatrix}e^{2t}$$

and the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}e^{2t} + c_2\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}te^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}e^{2t}\right)$$

$$+c_3\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\frac{t^2}{2}e^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}te^{2t} + \begin{bmatrix} 5 \\ 11 \\ 1 \end{bmatrix}e^{2t}\right)$$

© Krista King Math